16. It's not too hard to understand, in rough terms, how the Planck length crept into Klein's analysis. General relativity and quantum mechanics invoke three fundamental constants of nature: *c* (the velocity of light), *G* (the basic strength of the gravitational force) and ħ (Planck's constant describing the size of quantum effects). These three constants can be combined to produce a quantity with units of length: $(\hbar G/c^3)^{1/2}$, which, by definition, is the Planck length. After substituting the numerical values of the three constants, one finds the Planck length to be about 1.616×10^{-33} centimeters. Thus, unless a dimensionless number with value differing substantially from 1 should emerge from the theory—something that doesn't often happen in a simple, well-formulated physical theory—we expect the Planck length to be the characteristic size of lengths, such as the length of the curled-up spatial dimension. Nevertheless, do note that this does not rule out the possibility that dimensions can be larger than the Planck length, and in Chapter 13 we will see interesting recent work that has investigated this possibility vigorously.

17. Incorporating a particle with the electron's charge, and with its relatively tiny mass, proved a formidable challenge.

18. Note that the uniform symmetry requirement that we used in Chapter 8 to narrow down the shape of the universe was motivated by astronomical observations (such as those of the microwave background radiation) within the *three large dimensions.* These symmetry constraints have no bearing on the shape of the possible six tiny extra space dimensions. Figure 12.9a is based on an image created by Andrew Hanson.

19. You might wonder about whether there might not only be extra space dimensions, but also extra time dimensions. Researchers (such as Itzhak Bars at the University of Southern California) have investigated this possibility, and shown that it is at least possible to formulate theories with a second time dimension that seem to be physically reasonable. But whether this second time dimension is really on a par with the ordinary time dimension or is just a mathematical device has never been settled fully; the general feeling is more toward the latter than the former. By